

SLIDE 1: TITLE PAGE

KalaSetu AI: A Zero-Barrier AI E-Commerce Marketplace & Smart Cataloging Ecosystem for Marginalized Artisans

- **Problem Statement ID:** SIH26090
- **Problem Statement Title:** AI-Driven Market Linkage & Smart Cataloging App for Marginalized Artisans
- **Theme:** Heritage & Culture / Miscellaneous
- **PS Category:** Software
- **Target Beneficiaries:** 35.22 Lakh Handloom & Handicraft Workers in India (72% Female, <30% Rural Literacy)
- **Team ID:** *[Insert Your Team ID]*
- **Team Name:** *[Insert Your Team Name]*

SLIDE 2: IDEA TITLE & COMPLETE SOLUTION ARCHITECTURE

Idea: An AI Virtual Business Manager + Zero-Barrier Marketplace

1. The Core Problem (The Last-Mile Digitization Gap)

- **For Rural Artisans (Supply Side Failure):** 72% female with <30% digital literacy. Existing platforms (Amazon Karigar, GoCoop) require English fluency, professional studio equipment, tax paperwork, and complex cataloging forms. Middlemen extract 300% to 500% profit cuts.
- **For Urban Buyers (Demand Side Failure):** Buyers see trending crafts on Instagram Reels/Pinterest but cannot find authentic rural makers, negotiate custom pieces, or verify genuine handcrafts from cheap factory machine fakes.

2. Our Complete Solution (Starting Core + Advanced Suite)

- **CORE FEATURE 1 AI Photo Studio (Offline Background Removal):** Raw phone camera photo → On-device TFLite (U2Net) removes messy rural backgrounds and color-corrects to GeM/ONDC e-commerce standards with ₹0 cloud cost.
- **CORE FEATURE 2 Voice-to-Catalog Engine (Bhashini AI):** Artisan speaks in their mother tongue → Bhashini transcribes and translates 22 Indian languages → LLM generates professional SEO titles, product stories, and specifications.
- **CORE FEATURE 3 Explainable Cost-Plus Pricing ML:** Regression model trained on real Kaggle handmade datasets calculates a transparent, fair price (Raw Materials + Labor Hours + Market Average), preventing underpricing and building buyer trust.
- **CORE FEATURE 4 Instant Digital Storefront & ONDC Linkage:** Automatic storefront generation with 1-click export to ONDC network, WhatsApp catalog, and direct-to-consumer QR links.
- **EXPANDED USP 1 "Reels-to-Artisan" Visual Search:** Screenshot from Reels/Pinterest → MobileCLIP + Supabase vector search matches authentic inventory in <200ms.
- **EXPANDED USP 2 AI Room Decor Space Stager:** Upload room/wall photo → Gemini Vision suggests matching handmade decor.
- **EXPANDED USP 3 10s KalaReels Authenticity Feed:** Proof-of-making micro-videos destroying counterfeit machine fake doubts.

- **EXPANDED USP 4** **Spoken Dukandar Hisab-Kitab:** Voice-first daily business summary (material stock, revenue, orders).

SLIDE 3: TECHNICAL APPROACH & ZERO-COST ARCHITECTURE

End-to-End Technical Workflow (6-Step Pipeline)

- **1. Raw Capture:** Artisan snaps raw product photo, speaks voice note, or shoots 10s making video on basic Android phone (< 150 MB RAM).
- **2. Edge AI & Tool Calling:** On-device **Needle 3 (8–29 MB)** & **Jev** perform instant local field extraction and claim risk validation + TFLite/Tesseract WASM for offline slip OCR.
- **3. Voice-to-Listing: Browser Native STT (Web Speech API) + Bhashini AI (22 languages) + Sarvam AI** transcribes regional audio → generates bilingual e-commerce listings.
- **4. Micro-Texture Vision (VLM): Qwen2-VL-2B-Instruct (90.1% DocVQA) + PaliGemma-2-3B** on NVIDIA RTX 3050 CUDA (4GB VRAM) inspects micro-texture weave patterns, authentic motifs, and Pehchan artisan ID cards.
- **5. Reasoning & Pricing Brain: Google Gemma 2 (2B/9B) / Qwen2.5-3B** + Scikit-learn Cost-Plus ML Regression suggests fair retail and wholesale pricing.
- **6. Supabase Cloud BaaS & ONDC Beckn: Supabase (Free Tier)** manages auth, Postgres DB, storage buckets & `pgvector` for visual search + **ONDC Beckn Protocol & Leaflet OSM Geo-Fencing** around registered SFURTI clusters.

100% Zero-Cost Tech Stack Table

Component	Technology Used	Cost Model	Core Functionality
Edge AI & On-Device Tool Calling	Needle 3 (8–29 MB) + Jev + TFLite	₹0 (On-Device Edge)	Sub-50ms local tool calling, field extraction & claim validation
Speech & Voice AI	Browser Native STT + Bhashini AI + Sarvam	100% Free Govt AI / Native	Zero-latency streaming STT/TTS across 22 Indian regional languages
Micro-Texture Vision (VLM)	Qwen2-VL-2B-Instruct + PaliGemma-2-3B	₹0 Local RTX 3050 CUDA	90.1% DocVQA SOTA craft motif recognition, weave audit & receipt OCR

Reasoning Brain & Pricing	Gemma 2 (2B/9B) + Scikit-learn ML	₹0 Local Host Engine	Cost-plus pricing regression + luxury catalog storytelling
Database, Auth & Storage	Supabase (Free Tier BaaS)	₹0 Free Tier	User auth, relational DB, media storage & pgvector visual search
Mapping & Marketplace	Leaflet.js OSM + ONDC Beckn Protocol	100% Free Open-Source	SFURTI cluster geo-fencing (378 clusters) & ONDC network sync

SLIDE 4: FEASIBILITY & MARKET SIZING

Market Sizing (TAM / SAM / SOM)

64.66 Lakh
TAM (Total Artisans in India)

₹775.92 Cr
Annual Platform Opportunity

12.93 Lakh
SAM (20% Cluster Reach)

300 Units
SOM Year 1 (Jaipur Pilot)

Market Metric	Definition & Scope	Target Numbers	Projected Value
TAM	Total Handloom & Handicraft workers in India (Ministry of Textiles)	64.66 Lakh artisans	₹775.92 Crore / year
SAM	Priority craft states reachable through established cluster networks	12.93 Lakh artisans (20%)	₹155.16 Crore / year
SOM (Year 1)	Jaipur Pilot clusters: Sanganer, Bagru (Block Print) & Muhana (Blue Pottery)	300 Artisans	₹3.60 Lakh pilot test

Feasibility & Risk Mitigation Strategies

Potential Bottleneck	Engineering & Business Mitigation
Offline Rural Connectivity	Offline-first local caching; photos, voice notes, and pricing ML run on device and sync when online.
Illiteracy & Accounting	Voice Dukandar Hisab-Kitab speaks aloud daily materials consumed, gross earnings, and pending orders.
Pricing Mistrust	Explainable Cost-Plus AI shows transparent cost breakdown (Material + Labor + Margin) rather than black-box price.

**Unavailable Items from
Reels**

Bespoke Custom Order Pipeline: Buyers send custom RFQ with photo; matched artisans quote via voice with milestone escrow.

SLIDE 5: COMPETITIVE ADVANTAGE & IMPACT

Competitive Matrix: Why Legacy Platforms Fail

Feature Comparison	Amazon Karigar / Meesho	GoCoop / IndiaHandmade	KalaSetu AI (Our Platform)
Artisan Onboarding	High friction (English, GST, Tax Forms)	Manual approval & bureaucratic paperwork	Zero-Barrier (Voice Note + Phone Camera)
Photo & Cataloging	Requires white studio background	Manual data entry	Automated TFLite Studio + Bhashini Voice
Pricing Assistance	None (Artisans undercut each other)	None (Manual pricing)	Explainable Cost-Plus ML Regression
Proof of Authenticity	None (Machine copies undercut handmade)	Static textual certification	10s KalaReels (Behind-the-scenes video proof)
Buyer Discovery	Keyword search only	Basic category navigation	AI Visual Search (Reels) + AI Room Stager
Custom Commissioning	✗ Not supported	✗ Not supported	✓ Direct Custom RFQ with Escrow Protection
Artisan Accounting	Complex desktop seller dashboards	None	Spoken Daily "Hisab-Kitab" (Voice Briefing)

Social & Economic Impact

- **Direct Income Growth:** Bypasses predatory middlemen margins (300%–500%), passing **90%+ direct revenue** to the artisan.
- **Women Empowerment:** Directly uplifts the **25.46 Lakh female handloom workers** (72% of total sector) who operate from home.
- **Global Market Bridge:** Unlocks direct access to the **₹33,424 Crore** Indian handicraft export demand through ONDC integration.

SLIDE 6: RESEARCH AND AUTHORITATIVE REFERENCES

Primary Data Sources & Government Citations

- **Handloom Demographics & Gender Split:** *4th All India Handloom Census 2019–20*, Ministry of Textiles, Govt. of India (PIB Release ID: 1744040) — Documents 35.22 Lakh handloom workers, 72% female participation, and average household craft earnings.
- **Handicrafts Export & Market Valuation:** *Export Promotion Council for Handicrafts (EPCH) 38th Annual Report 2024-25* — Validates ₹33,424 Crore export volume and global demand for authentic Indian GI crafts.
- **Cluster-First Onboarding Network:** *Ministry of MSME SFURTI (Scheme of Fund for Regeneration of Traditional Industries) Annual Directory 2024* — Documents 378 active artisan clusters for targeted pilot onboarding.
- **National Digital Infrastructure:** *Open Network for Digital Commerce (ONDC) Retail & Logistics Architecture Whitepaper* (ondc.org) — Open protocols for logistics integration.
- **National Language Translation Engine:** *Bhashini AI Mission*, Digital India Bhasha Kendra, Ministry of Electronics & IT (MeitY) — 22 Scheduled Indian language speech recognition APIs.
- **Pricing ML Training Datasets:** *Kaggle Handmade Products & Indo Fashion Datasets* — Multi-feature regression for fair cost-plus pricing models.
- **Computer Vision & Vector Benchmarks:** *OpenAI CLIP & Apple MobileCLIP Research Papers* (2024) — Validates sub-200ms mobile visual search and image embedding accuracy.